When the Machine Stops Looking: Autonomous Replenishment, Censored Demand, and the Vanishing Value of a Human Filter

Abstract

Retailers increasingly delegate replenishment to algorithms that learn from sales with no human in between. A learner can correct for demand censoring on its own when stockouts are recorded on clean days, but it cannot flag contaminated records without domain knowledge or access to the returns-and-corrections log. We study a censoring-aware but contamination-blind learner in a repeated newsvendor in which a fraction of high-demand events is masked by returns or corrections booked to those periods, making them appear as low-demand days. The order and the data then form a closed loop. We show that the loop drives the order to a unique equilibrium below the optimum, that no tipping point arises along the way, and that the marginal service loss per unit of contamination equals the item's stockout frequency $1-\beta$. The items a firm automates first, namely low-margin, high-volume, thin-service items, degrade fastest. The loss accrues every period, so the value of a human who screens the data grows with the planning horizon and is concentrated in screening the data rather than in overriding orders, which matches the field evidence on human discretion. A minimal filter that audits only the returns-and-corrections log recovers nearly first-best service. We validate the equilibrium, the exposure law, and the filter on a calibrated instance.

1 Introduction

Large retailers now let software place replenishment orders with no buyer in between. The system reads recent sales, updates its view of demand, and sets the order quantity q_t for each of hundreds of thousands of items, night after night. A buyer used to do two things that the order quantity alone does not record. She read a sold-out day as understating true demand, and she set aside the records she judged unreliable, that is, the week a promotion, a returns batch, or a feed error distorted the numbers. The first correction is mechanical when labels are clean. The firm knows when an item stocked out, so a machine can make it. The second correction needs to know the business, and it matters most when returns, phantom inventory, or feed errors mask a stockout in the feed. This paper asks what the firm loses when the second correction goes away, and shows that for a recognizable class of items the loss is not a steady efficiency tax but a service level that drifts down and settles well below target.

Consider an item the firm wants stocked to a fifty percent service level, so that the cost-minimizing order is the median of demand, namely 100 units. Suppose that on a tenth of the high-demand events, a returns batch or data correction is booked to the same period, reducing the recorded demand so that what should be recorded as a stockout instead appears as an ordinary low-sales day, say around 40 units on average. A censoring-aware learner corrects for the stockouts it can see, yet it reads these masked days at face value as evidence that demand is weak, so it trims the order. The lower order produces more high-demand events, more of which can be masked by coincidental returns or corrections, and the order keeps drifting down until it settles near 96 units at a realized service level of forty-four percent. Raise the masking rate to thirty percent of high-demand days and the same item settles near 83 units at twenty-nine percent service, and a low-service item that started at thirty percent settles in single digits. The two clean alternatives both fail. Full automation is cheap per decision and reacts overnight, but it walks the item down. Full human review corrects every record by checking the returns log and removing the booked offsets, but it spends scarce attention on every item the firm carries, and the field evidence is that letting buyers also override the order itself reduces profit (Kesavan and Kushwaha, 2020). The question is where between these extremes the firm should sit, and the answer turns on how much of the high-demand events are masked by returns and on how exposed the item is to the loop.

The difficulty is that the order and the data form a closed loop. Because stockouts are recorded on clean days, one might expect a censoring-aware learner to be safe, and on clean records it is. The catch is that the de-censoring correction trusts the stockout labels, and contamination masks exactly those labels. A masked stockout enters the learner's data as a genuine low-demand day, lowering the order, and the lower order raises the share of days that stock out and so can be masked. The learner cannot separate a genuinely weak demand signal from a corrupted one, because the evidence that would separate them, namely the demand it failed to serve, never appears in the feed. Saying where this loop settles, and how fast it descends, means tracking a decision-dependent estimator together with the order it feeds, and neither piece can be analyzed in isolation.

The closest prior work studies learning under censoring on clean data. Besbes and Muharremoglu (2013) isolate the regret cost of censoring, Lariviere and Porteus (1999) show that ordering above the myopic level keeps the upper tail visible and escapes the censoring trap, and Ding et al. (2024) build censoring-aware learners that attain vanishing regret with features. Hssaine and Sinclair (2024) make the observable boundary central, but in an offline setting with uncorrupted records. Mersereau (2015) show that inventory record inaccuracy compounds the downward bias already present under censoring. Each of these assumes the records are clean, or treats inaccuracy separately from the learning loop, which is the one correction a machine cannot make on its own, because contamination, unlike censoring, is not self-revealing in the demand feed. So the question this paper asks is the following. When a censoring-aware but contamination-blind learner runs unattended, does data contamination wash out as noise, or does it compound through the censoring loop into a self-reinforcing decline, and how little human screening of the data is enough to stop it? Answering it requires coupling a contamination model with an endogenous observable boundary inside a repeated newsvendor, and then pricing a scarce human filter against the loop.

Contamination does not wash out. It compounds. The order descends to a single equilibrium below the target and stays there, and the descent is smooth, so there is no warning tipping point to watch for. What governs the damage is one number. The service the firm loses per unit of contamination equals the item's stockout frequency, namely one minus its target service level. Low-service, thin-margin, high-volume items run with high stockout frequencies, so they bleed service fastest, and they are the items a firm automates first. Because the shortfall repeats every period, the cost of leaving the loop unattended grows with the planning horizon rather than washing out, and the value of a human sits entirely in screening the data rather than in overriding the order, which is why input-level human corrections help in the field while order overrides do not (Kesavan et al., 2025). The remedy is cheap and pointed. Masked stockouts leave a trace in the returns-and-corrections log, so a human called in only on logged correction days audits exactly the records at fault, and because the cost penalty grows with the square of the contamination, auditing even half of those days removes most of it while auditing all of them restores the target. The contributions below make each of these statements precise and validate them on a calibrated instance.

Contributions.

1. **Model.** We formulate the repeated newsvendor as a closed loop in which a censoring-aware but contamination-blind learner sets orders, demand is censored at the order, and a fraction ϵ of true stockouts is masked in the feed as low-sales days drawn from a downward source G, following the returns-and-inventory-inaccuracy patterns documented by Mersereau (2015); DeHoratius et al. (2008). Lemma 1 derives the population estimate the learner converges to, namely $\widehat{F}_q(x) = F(x) + \epsilon(1 - F(q))G(x)$, in which the contamination term carries the stockout frequency $1 - F(q)$ and so depends on the order.

2. **Main result, with a negative result inside it.** Theorem 1 is the spine of the paper. Part (i) proves that the projected Robbins–Monro learner converges almost surely to a unique equilibrium $q_\infty(\epsilon, \beta) < q^\star$ and that the order map is monotone, so no tipping point or bistable collapse can arise from this contamination. Part (ii) is the centerpiece, the exposure law $\mathrm{d}\,[\text{service}]/\mathrm{d}\epsilon = -(1-\beta)G(q^\star)$, equal to $-(1-\beta)$ in the relevant range, so degradation is steepest for low-service items. Part (iii) shows long-run average cost exceeds the optimum by $\Theta\big((b+h)\,\epsilon^2(1-\beta)^2/f(q^\star)\big)$.

3. **Prescription.** Because that penalty is quadratic in contamination, a correction-log alarm that audits a fraction φ of logged correction days removes more than three quarters of it at half coverage and restores the optimum at full coverage (Theorem 1(iii), Table 1). Oversight should be triggered by the returns and corrections log rather than by stockout flags, and its return is concentrated at the data-screening level rather than the order-override level, which rationalizes the opposing signs of the two field experiments on human discretion (Kesavan et al., 2025; Kesavan and Kushwaha, 2020).

4. **Numerical evidence.** On a calibrated instance with normal demand we confirm the unique equilibrium, the exact exposure law, and the simulated order path against the mean-field prediction across critical ratios, contamination levels, and horizons, and we report the items most exposed to the loop (Section 5).

Section 2 places the result in the literature. Section 3 states the model and the learner. Section 4 derives the equilibrium, the exposure law, and the value of the filter. Section 5 reports the numerical study.

2 Related Literature

Our work draws on six strands, and we keep to the top journals in each.

Information stalking and the exploration trap. A censored learner that only exploits under-orders and so never observes the upper tail, converging to a level below the optimum (Lariviere and Porteus, 1999; Ding et al., 2002). The known remedy is to order above the myopic level to acquire information. Our equilibrium looks similar but is driven by contamination rather than by myopia, and we show that the remedy of de-censoring does not remove it, because masked stockouts suppress the labels the de-censoring relies on.

Adaptive learning and regret under censoring. A mature line designs censoring-aware policies with vanishing regret, through distribution-free stochastic-gradient learning (Godfrey and Powell, 2001; Huh and Rusmevichientong, 2009), the Kaplan–Meier estimator (Huh et al., 2011), the isolation of the regret cost of censoring (Besbes and Muharremoglu, 2013), inference for the resulting policies (Ban, 2020), and the recent offline analysis of Hssaine and Sinclair (2024) that makes the observable boundary explicit. We adopt this regret language and study the realistic learner that is censoring-aware yet contamination-blind.

Spiral-down and decision-dependent demand. Cooper et al. (2006) formalize the loop in which censored observations feed forecasts that lower controls that further censor observations. Our descent is a structured instance of performative prediction, namely the setting where deploying a model shifts the distribution it then faces (Perdomo et al., 2020). We give the operations form, derive its equilibrium in closed form, and tie it to oversight design.

Data-driven, machine learning, and reinforcement learning newsvendor. The feature-based data-driven newsvendor (Ban and Rudin, 2019), prescriptive analytics (Bertsimas and Kallus, 2020; Elmachtoub and Grigas, 2022), censoring-aware feature-based control (Ding et al., 2024), learning under shifting demand (Chen, 2021), and deep reinforcement learning for lost-sales inventory (Gijsbrechts et al., 2022) all engineer performance under censoring. Our point is orthogonal. Even a well-engineered censoring-aware learner is driven below target by masked stockouts absent a human filter.

Inventory record inaccuracy and masked stockouts. DeHoratius and Raman (2008); DeHoratius et al. (2008) document that retail systems often carry phantom inventory, that is, positive on-hand records when the shelf is empty. Mersereau (2015); Chen and Mersereau (2015) show that this inaccuracy compounds the downward bias already present under demand censoring. We route the same force through a repeated learning loop in which the identification boundary is the order itself, and tie the remedy to the returns-and-corrections log that records when offsets are booked (Kesavan et al., 2025).

Robust and distribution-free inventory. The contamination model is in the spirit of Huber's (1964) ϵ-contamination, the classical robustness device, and connects to distribution-free inventory in the tradition of Scarf (1958). We are not aware of prior work routing contamination through a censored, repeated inventory loop in which the identification boundary is itself a decision.

Human-in-the-loop and algorithm aversion. Field experiments show that human override helps at the data-input level and hurts at the decision level (Kesavan et al., 2025; Kesavan and

Kushwaha, 2020), a behavioral literature documents algorithm aversion and how modest control over the algorithm restores adoption (Dietvorst et al., 2018), and related work asks how to elicit and combine human judgment (Ibrahim et al., 2021). We give a model in which the human's value is precisely as a data filter, which rations the scarce attention these papers measure.

3 Model and the Autonomous Learner

Demand D_t is drawn i.i.d. from a continuous, strictly increasing distribution F with density f. The newsvendor underage and overage costs are b and h, the critical ratio is $\beta = b/(b+h)$, and the cost-minimizing order is the β-quantile $q^\star = F^{-1}(\beta)$. Each period the firm orders q_t and observes sales and a stockout label.

Contamination through returns and corrections. In each period, returns and data corrections are logged as an operational byproduct, independent of demand. With intensity ϵ, a true high-demand event (demand $D_t \geq q_t$ that should be recorded as a stockout) coincides with a return or correction being booked to that period, reducing the recorded demand and masking the stockout. Specifically, the recorded demand becomes a value drawn from a downward source G, where G is stochastically dominated by F and represents the magnitude of typical returns or corrections. The recorded value and the operative stockout status are now decoupled, and the stockout label follows the recorded value rather than the true demand. This is the one error a machine cannot correct on its own without domain knowledge or access to the returns and corrections log, which contains the ground truth for when this offset occurred.

Recorded data. Work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with a filtration $(\mathcal{F}_t)$, and let the order q_t be $\mathcal{F}_{t-1}$-measurable. In period t, given q_t, nature draws the demand $D_t \sim F$, a contamination coin $U_t \sim \mathrm{Unif}[0,1]$, and if contamination occurs, a correction value $D_t^G \sim G$, mutually independent and independent of $\mathcal{F}_{t-1}$. The recorded outcome is one of three. If $D_t < q_t$ the day is served and recorded at value D_t. If $D_t \geq q_t$ and $U_t > \epsilon$ the day is a genuine high-demand event, recorded as censored at q_t (stockout). If $D_t \geq q_t$ and $U_t \leq \epsilon$ a correction is booked for that period, reducing recorded demand to D_t^G, which enters the learner's data as an uncensored low-sales observation. The returns-and-corrections log records which periods experienced a booking; call this a *logged correction day*. Let $g_t = \mathbf{1}\{\text{recorded demand} < q_t\}$ denote the recorded served-indicator, which is $\mathcal{F}_t$-measurable.

The learner and the audit. The autonomous policy is censoring-aware and contamination-blind. It uses the stockout labels, so it scores a genuine high-demand event as demand above the order, but it cannot tell a masked high-demand day from a true low-sales day without access to the returns-and-corrections log. We analyze the canonical adaptive form, a projected Robbins–Monro recursion that drives the recorded served-rate to the target β,

$$q_{t+1} = \Pi_{[\underline{q},\overline{q}]}\big(q_t - a_t\,(g_t - \beta)\big), \qquad a_t > 0, \quad \textstyle\sum_t a_t = \infty, \quad \sum_t a_t^2 < \infty, \tag{1}$$

where Π is the projection onto a compact interval $[\underline{q}, \overline{q}]$ that contains $q^\star$ in its interior. The estimate-then-optimize learner that de-censors by the Kaplan–Meier estimator over a rolling window and orders the β-quantile of its estimate is an alternative with the same mean field, by Lemma 1, and is the version we simulate in Section 5. A human filter operates at the data level by consulting the returns-and-corrections log and, for a fraction φ of the logged corrections, undoing the recorded offset to recover the true demand signal and feeding it back into the learner's data stream. This data-level human intervention is shown to improve profit in the field, in contrast to order-level overrides which reduce it (Kesavan et al., 2025).

4 The Contamination Spiral and the Exposure Law

We now ask where the loop settles, how fast it descends, and what restores it, keeping the clean benchmark $q^\star$ in view throughout. The first step records what the learner perceives at a fixed order and reduces the recursion to a one-dimensional fixed-point problem.

Lemma 1 (Perceived demand and the mean field)**.** *Fix an order* $q \in [\underline{q}, \overline{q}]$. (a) *Operating persistently at* q*, the estimate-then-optimize learner's windowed estimate converges almost surely and uniformly to*

$$\widehat{F}_q(x) = F(x) + \epsilon\,(1 - F(q))\,G(x), \qquad x \le q. \tag{2}$$

(b) *The recorded served-indicator satisfies* $\mathbb{E}[g_t \mid q_t = q] = \widehat{F}_q(q) = H(q)$*, where*

$$H(q) \;:=\; F(q) + \epsilon\,(1 - F(q))\,G(q). \tag{3}$$

Proof. (a) At the fixed level q the recorded values are i.i.d., and with a single censoring level the Kaplan–Meier estimator equals the empirical distribution on $[0, q]$ with the censored mass placed above q. The uncensored records at or below $x < q$ are the genuine demands below x, of probability $F(x)$, together with the masked stockouts whose correction value falls below x, of probability $\epsilon(1 - F(q))G(x)$. Glivenko–Cantelli gives almost sure uniform convergence to their sum (2). (b) The indicator g_t equals one on exactly the served and masked-below days, so $\mathbb{E}[g_t \mid q_t = q] = F(q) + \epsilon(1 - F(q))G(q) = \widehat{F}_q(q)$. □

The genuine uncensored days below x contribute $F(x)$. The masked stockout days contribute the second term, in which $\epsilon(1 - F(q))$ is the share of days that stock out and are masked in the feed. The factor $1 - F(q)$ is the stockout frequency, and it ties the size of the contamination to the order itself. That single coupling is what turns a static bias into a loop, because the equilibrium order solves

$$H(q) = \beta. \tag{4}$$

The whole paper rests on the following result, which reads the loop in three parts, namely where it goes, how much it costs at the margin, and how cheaply a human reverses it.

Theorem 1 (Contamination spiral and the exposure law)**.** *Let* F *be continuous and strictly increasing with* $q^\star = F^{-1}(\beta)$*,* $\beta \in (0, 1)$*, and let* G *be stochastically dominated by* F*. Under the censoring-aware, contamination-blind learner of Section 3 with contamination intensity* ϵ*, the following hold.*

(i) (Spiral to a unique equilibrium, no tipping point) *The map* H *is strictly increasing, so* (4) *has a unique root* $q_\infty(\epsilon, \beta)$*, and the projected recursion* (1) *converges to it almost surely* *from any start in $[\underline{q}, \overline{q}]$. The root satisfies $q_\infty = q^\star$ at $\epsilon = 0$ and $q_\infty < q^\star$, strictly decreasing in ϵ, for $\epsilon > 0$. No second equilibrium or tipping point exists.*

(ii) (Exposure law) *The equilibrium service level $F(q_\infty)$ is smooth and strictly decreasing in ϵ, with*

$$\left.\frac{\mathrm{d}\, F(q_\infty)}{\mathrm{d}\epsilon}\right|_{\epsilon=0} = -(1-\beta)\, G(q^\star), \tag{5}$$

which equals $-(1-\beta)$ when the contamination lies below $q^\star$. The marginal service loss is the item's stockout frequency.

(iii) (Cost and oversight) *Long-run average cost exceeds the optimum by*

$$C(q_\infty) - C(q^\star) = \tfrac{1}{2}(b+h) f(q^\star)\,(q^\star - q_\infty)^2 + o\big((q^\star - q_\infty)^2\big) = \Theta\Big(\tfrac{(b+h)\,\epsilon^2 (1-\beta)^2}{f(q^\star)}\Big). \tag{6}$$

A correction-log alarm that audits a fraction φ of logged correction days corrects the masked records at that rate, reducing the effective contamination to $\epsilon(1-\varphi)$ and the cost penalty to a $(1-\varphi)^2$ fraction, so half coverage removes three quarters of the penalty and full coverage restores $q^\star$. The value of oversight is largest for low-service items.

Proof of Theorem 1. Part (i), structure. Differentiating (3), $H'(q) = f(q)\,(1-\epsilon G(q)) + \epsilon\,(1-F(q))\,g(q)$, and both terms are nonnegative because $\epsilon G(q) \le 1$, with the first strictly positive, so H is strictly increasing and the root of (4) is unique. The added term in (3) is positive for $\epsilon > 0$, so $F(q_\infty) < \beta$, that is $q_\infty < q^\star$, and monotonicity in ϵ follows from the implicit function theorem.

Part (i), convergence. Write $h(q) = \beta - H(q)$. By Lemma 1(b) the recursion (1) is $q_{t+1} = \Pi_{[\underline{q},\overline{q}]}(q_t + a_t h(q_t) + a_t M_t)$ with $M_t = \mathbb{E}[g_t \mid \mathcal{F}_{t-1}] - g_t$, a martingale-difference sequence with $|M_t| \le 1$. Strict monotonicity of H gives the Lyapunov sign condition $(q - q_\infty)h(q) < 0$ for every $q \neq q_\infty$ in $[\underline{q}, \overline{q}]$. Set $V_t = (q_t - q_\infty)^2$. Since q_∞ lies in the interior and Π is nonexpansive with $\Pi(q_\infty) = q_\infty$,

$$\mathbb{E}[V_{t+1} \mid \mathcal{F}_{t-1}] \;\le\; V_t + 2a_t\,(q_t - q_\infty)\,h(q_t) + a_t^2\,\mathbb{E}[M_t^2 \mid \mathcal{F}_{t-1}] \;\le\; V_t - 2a_t\,\Delta_t + a_t^2,$$

where $\Delta_t = (q_t - q_\infty)(H(q_t) - \beta) \ge 0$. As $\sum_t a_t^2 < \infty$, the Robbins–Siegmund nonnegative almost-supermartingale theorem (Robbins and Siegmund, 1971) implies that V_t converges almost surely and $\sum_t a_t \Delta_t < \infty$ almost surely. Were $|q_t - q_\infty| \ge \delta$ for all large t on a positive-probability set, then Δ_t would be bounded below by $\inf_{\delta \le |q - q_\infty|} \Delta(q) > 0$ there, forcing $\sum_t a_t \Delta_t = \infty$ because $\sum_t a_t = \infty$, a contradiction. Hence $\liminf_t |q_t - q_\infty| = 0$, and since V_t converges, $q_t \to q_\infty$ almost surely (Robbins and Monro, 1951; Kushner and Yin, 2003). This holds from any start in $[\underline{q}, \overline{q}]$, so there is no second equilibrium and no tipping point.

Part (ii). Differentiate $H(q_\infty(\epsilon)) = \beta$ in ϵ at $\epsilon = 0$, where $q_\infty = q^\star$ and $F(q^\star) = \beta$. The cross terms in ϵ vanish, leaving $f(q^\star)\, q'_\infty(0) + (1-\beta)\, G(q^\star) = 0$, and multiplying by $f(q^\star)$ gives (5). Part (iii). Since $C'(q) = (b+h)F(q) - b$, we have $C'(q^\star) = 0$ and $C''(q^\star) = (b+h)f(q^\star)$, which gives the quadratic expansion. Substituting $q^\star - q_\infty = \epsilon(1-\beta)G(q^\star)/f(q^\star) + o(\epsilon)$ from (5) gives the order term. A correction-log alarm reviews a fraction φ of logged correction days. Each masked stockout appears in that log, so an audited day is corrected with probability φ, the effective intensity is $\epsilon(1-\varphi)$, and the cost, quadratic in intensity by (6), scales as $(1-\varphi)^2$. $\square$

The same structure pins down the rate, and with it the cumulative cost behind the $\Theta(T)$ claim. The step size enters through the local slope $\lambda := H'(q_\infty) > 0$ established in part (i).

Lemma 2 (Rate)**.** *Under* (1) *with* $a_t = a/t$ *and* $2a\lambda > 1$, $\mathbb{E}[(q_t - q_\infty)^2] = O(1/t)$ *and* $\big|\mathbb{E}[q_t - q_\infty]\big| = O(1/t)$.

This is the classical stochastic-approximation rate for a recursion whose mean field is locally linear and strictly stable at its root (Kushner and Yin, 2003). It controls the transient and isolates the contamination bias $q^\star - q_\infty$ as the term that persists.

Lemma 3 (Cumulative regret)**.** *Let* $R_T = \sum_{t=1}^{T}\big(\mathbb{E}[C(q_t)] - C(q^\star)\big)$ *under the step size of Lemma 2. Then*

$$R_T = T\,\big(C(q_\infty) - C(q^\star)\big) + O(\log T) = \Theta\Big(\tfrac{(b+h)\,\epsilon^2(1-\beta)^2}{f(q^\star)}\,T\Big).$$

Proof. A second-order expansion gives $C(q_t) - C(q_\infty) = C'(q_\infty)(q_t - q_\infty) + \frac{1}{2}C''(\zeta_t)(q_t - q_\infty)^2$ for some ζ_t between q_t and q_∞, with C'' bounded on $[\underline{q}, \overline{q}]$. Taking expectations and applying Lemma 2 to both terms gives $\mathbb{E}[C(q_t)] - C(q_\infty) = O(1/t)$, which sums to $O(\log T)$. Adding $T\,(C(q_\infty) - C(q^\star))$ and substituting the constant order from (6) gives the claim. □

Theorem 1 captures an item run unattended on censored, contaminated data, and the mechanism is the factor $1 - F(q)$ in (2). The learner cannot separate a genuinely weak demand signal from a corrupted one, its censoring-aware response to an apparently weak signal is to order less, and ordering less raises the stockout frequency that feeds the next period more corrupted records. Two readings follow. First, the descent has no warning sign, because H is monotone and there is no second equilibrium, so a manager watching for a sudden break will see none and the shortfall arrives instead as a slow, permanent erosion. Second, the marginal damage is the stockout frequency $1 - \beta$, so a firm can rank its catalogue by target service level with data already in hand and find that the cheap, fast-moving, thin-service items it automates first are exactly the items the loop punishes hardest. The failure mode to remember is that de-censoring does not protect the item and a better optimizer does not help, because the data are the problem. The complementary priority is a human who screens logged correction days where the masking enters, and because the penalty is quadratic in contamination, auditing even half of those days removes most of it.

5 Numerical Study

We calibrate demand as $F = \mathcal{N}(100, 30)$ and the contamination source as $G = \mathcal{N}(40, 10)$, that is returns batches booked at about 40 units. We simulate the learner with $N = 200$ demand draws per period, a rolling window of $W = 60$ periods, horizon $T = 4000$, and a generous start of $q_0 = 160$, and we add a correction-log alarm that audits a fraction φ of logged correction days. Table 1 reports the long-run order against the mean-field equilibrium, the realized service, the cost gap above the optimum, and the audit load, namely the fraction of item-days the human inspects.

We simulate the estimate-then-optimize learner, which shares the mean field of the analyzed Robbins–Monro recursion by Lemma 1. The Robbins–Monro learner of (1) converges to the same equilibria, matching the roots of (4) to within 0.2 units across the cases below.

The table confirms every part of Theorem 1. The unfiltered learner ($\varphi = 0$) starts well above the optimum and descends to the mean-field equilibrium to within about a unit, validating part (i) and the absence of a tipping point. The damage is large and ranked by service level as part (ii) predicts. The low-service item at $\beta = 0.30$ loses almost all of its service and runs 46 percent above

Table 1: Simulated learner and correction-log filter. Demand $\mathcal{N}(100, 30)$, contamination $G = \mathcal{N}(40, 10)$ on masked stockout days, $\epsilon = 0.30$ for the contaminated rows. φ is the fraction of logged correction days audited. The optimum is $q^\star = 84.3$ for $\beta = 0.30$ and $q^\star = 115.7$ for $\beta = 0.70$. Cost gap is percent above optimal newsvendor cost. Audit load is the fraction of item-days inspected. Standard deviations are over the final 500 periods.

β	ϵ	φ	mean-field q_∞	sim q_∞ (s.d.)	service	cost gap %	audit load
0.30	0.00	–	84.3	83.2 (0.02)	0.287	0.1	0.000
0.30	0.30	0.0	51.3	51.6 (0.61)	0.053	45.7	0.000
0.30	0.30	0.5	72.1	72.5 (0.65)	0.180	7.1	0.122
0.30	0.30	1.0	84.3	83.1 (0.04)	0.287	0.1	0.212
0.70	0.00	–	115.7	114.2 (0.02)	0.682	0.1	0.000
0.70	0.30	0.0	105.4	104.2 (0.03)	0.555	7.8	0.000
0.70	0.30	0.5	111.3	110.1 (0.03)	0.631	1.8	0.054
0.70	0.30	1.0	115.7	114.3 (0.04)	0.683	0.1	0.093

optimal cost, while the high-service item at $\beta = 0.70$ loses far less. The filter validates part (iii) and its convex recovery. Auditing half of the logged correction days removes more than three quarters of the cost penalty, from 45.7 to 7.1 percent at $\beta = 0.30$ and from 7.8 to 1.8 percent at $\beta = 0.70$, and full coverage restores the optimum. The audit load shows the trade-off the manager faces. Because the alarm fires only on logged correction days, namely the masked stockouts themselves, full recovery requires reviewing about $\epsilon(1 - F(q_\infty))$ of item-days, far less than the stockout rate the old stockout-triggered alarm would have implied, and half coverage already captures most of the value.

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